

Aaron Zoll

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EDUCATION

Johns Hopkins, Graduate — Ph.D in Applied Mathematics	August 2023 – Present
Johns Hopkins, Graduate — M.S in Applied Mathematics	August 2022 – May 2023
Johns Hopkins, Undergraduate — B.S in Applied Mathematics	August 2019 – May 2022
Johns Hopkins, Undergraduate — B.A in Mathematics	August 2019 – May 2022

TEACHING EXPERIENCE

Johns Hopkins, Baltimore — Course Instructor Summer 2022 – Present

Developed a complete syllabus, lecture notes, and assignments to facilitate courses ranging from 15 to 80 students. Presented extensive lectures on linear algebra, probability, statistics, data analysis, and differential equations. Facilitated student engagement and discussion.

Johns Hopkins, Baltimore — Directed Reading Program Mentor Spring 2025/Fall 2025

Paired with up to three students a semester, mentoring a self-contained course on gradient descent, its convergence theory and guarantees. Wrote a series of lecture notes and exercises for students who were expected to discuss and work independently for a few hours every week. After the semester of preparation, each student presented a structured talk to the group of mentees.

Johns Hopkins, Baltimore — Course Developer Summer 2021

Contributed to the design and development of new mathematics and engineering courses. Authored original lecture material, guided Excel practice problems, engaging problem sets, and instructional resources tailored to enhance pedagogical clarity and student engagement.

Johns Hopkins, Baltimore — Teaching Assistant January 2020 – Present

Supported instruction in twenty undergraduate and graduate-level courses through grading, writing lecture notes, designing assignments, leading weekly discussion sections, and providing academic support to students. Collaborated with faculty to reinforce core course concepts and foster a strong learning environment.

Johns Hopkins, Baltimore — MSE Orientation Review Session August 2023/2024

Designed and led a series of review sessions for the incoming 120 masters students, covering foundational concepts in linear algebra and matrix analysis. Developed comprehensive lecture notes and facilitated interactive discussions to prepare students for rigorous graduate coursework.

Mathnasium, Baltimore — Math Instructor August 2019 – December 2019

Tutored dozens of students ages 6 to 18 in fundamental math topics ranging from multiplication tables to AP calculus. Adapted instruction to individual learning styles, promoting confidence and mastery in mathematical skills.

COURSE DEVELOPMENT

College Algebra (AS.110.102) — Complete Summer 2021

Johns Hopkins University

Collaborated with the Director of Online Programs to develop a comprehensive College Algebra course aimed at preparing incoming students for success in higher-level mathematics. Designed instructional content to reinforce key algebraic concepts through accessible and engaging materials.

Data Analysis Workshop (AS.110.100) — Complete

Summer 2022

Johns Hopkins University

Developed and launched a summer course for high school students introducing the fundamentals of data analysis, probability, and statistics. Encouraged students to master affective presentation skills and collaborative work. Produced a full suite of materials, including lecture videos, online quizzes, interactive assignments, and guided Excel tutorials, delivered to over 50 students annually.

INSTRUCTOR OF RECORD

Data Analysis Workshop (AS.110.100)

Summer 2023–2026

Johns Hopkins University

In this two-week pre-college program, students work in groups to construct and present a data analysis project which collects, organizes, cleanses, and visualizes a dataset of their choosing. Topics include exploratory data analysis, data visualization, probability distributions, data scraping and cleansing, the basics of hypothesis testing, and regression modeling.

Linear Algebra and Differential Equations (EN.553.291)

Fall 2026

Johns Hopkins University

An introduction to the basic concepts of linear algebra, matrix theory, and differential equations that are used widely in modern engineering and science.

TEACHING ASSISTANT

Graduate Teaching Assistant

Introduction to Computational Mathematics (EN.553.385)	Spring 2025, Spring 2026
Machine Learning 1 (EN.553.740)	Fall 2025
Introduction to Convexity (EN.553.665)	Fall 2024, Spring 2024
Matrix Analysis and Linear Algebra (EN.553.792)	Fall 2023
Mathematical Game Theory (EN.553.653)	Spring 2023
Mathematical Modeling and Consulting (EN.553.400)	Spring 2023
Optimization in Finance (EN.553.661)	Fall 2022

Undergraduate Teaching Assistant

Real Analysis I (EN.553.405)	Spring 2022, Summer 2022
Cryptology and Coding (EN.553.371)	Spring 2022
Calculus II (For Biological and Social Science) (AS.110.107)	Spring 2022
Honors Discrete Mathematics (EN.553.172)	Fall 2021
Differential Equations and Applications (AS.110.302)	Fall 2021, Spring 2021
Discrete Mathematics (EN.553.171)	Spring 2021, Fall 2020
Calculus III (AS.110.202)	Fall 2020
Introduction to Computing (AS.205.205)	Spring 2020

RESEARCH

Optimization Research

Ongoing

Dissertation research with Dr. Benjamin Grimmer, Johns Hopkins University

Conducting theoretical research on algorithm design and analysis to unify the regimes between smooth and nonsmooth convex problem classes (e.g. function exhibiting Hölder smoothness or uniform convexity). Prior work focused on heterogeneously smooth and convex compositions, calculus

results expanding and characterizing dual notations between Hölder smoothness and uniform convexity, interpolation theory for inexactly smooth convex functions, performance estimation over respective problem classes, and universal algorithm design. Future work entails characterizing the class of minimax optimal methods for convex Lipschitz minimization.

AI Pedagogy Research

Ongoing

Extracurricular research led by Dr. Sergey Kushnarev, Johns Hopkins University

Experimentation and implementation of various AI derived course assistants. Herein, we discuss, analyze, and test different methods for embedding the rising LLM tools into the education system. By supplying agents with structured prompts, focused on providing students with motivation and step-by-step guidance instead of direct answers, we aim to improve comprehension and intellectual capability.

Signal Processing Research

Upcoming

Planned collaboration with Dr. Mario Michelli & Kaleigh Rudge, Johns Hopkins University

Preparing to investigate spectral properties of the Discrete Fourier Transform and its connections to signal representation and harmonic analysis. Further investigation will include advancing understanding of the Fractional Fourier Transform, smoothly interpolating between the signal and frequency domains.

PUBLICATIONS

Accepted Papers

A Universally Optimal Primal-Dual Method for Minimizing Heterogeneous Compositions

Aaron Zoll, Benjamin Grimmer

IMA Journal of Numerical Analysis, 2025

arXiv:2503.07566

Preprints

A Complete Characterization of Optimal Subgradient Methods for Lipschitz Minimization

Aaron Zoll, Benjamin Grimmer

2026

arXiv:2607.19240

Inexactly Smooth Performance Estimation and New Optimized Gradient Methods

Aaron Zoll, Benjamin Grimmer

2026

arXiv:2606.01505

TALKS

On Interpolation Theory

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Mar, 2026

Algebra and Number Theory Guest Lecture, *Clayton High School*

Apr, 2026

On Inexactly Smooth Performance Estimation

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Sept, 2025

INFORMS Annual Meeting, *Atlanta, Georgia*

Oct, 2025

Junior MINDS Seminar, Johns Hopkins Mathematical Institute for Data Science

Nov, 2025

Modeling and Optimization: Theory and Applications (MOPTA), *Lehigh University*

(Upcoming) Aug, 2026

On Minimizing Heterogeneous Compositions

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Mar, 2025

Junior MINDS Seminar, Johns Hopkins Mathematical Institute for Data Science

Apr, 2025

International Conference on Continuous Optimization (ICCOPT), *Los Angeles, California*

Aug, 2025

AWARDS

Joel Dean Award for Excellence in Teaching, department of Applied Math and Statistics (2019-2020)

Joel Dean Award for Excellence in Teaching, department of Mathematics (2019-2020)

Gordon L. and Beatrice C. Bowles Fellowship (2022-2023)

Promotion to Teaching Fellow in the Department of Applied Mathematics and Statistics, Johns Hopkins University (2025)

SKILLS

Expertise in mathematical problem-solving, analytical reasoning, and quantitative analysis, with the adept ability to tackle complex challenges.

Aptitude for delivering clear, engaging presentations and cultivating a dynamic, intellectually stimulating classroom environment.

Proficiency in Matlab, Python, and Julia, with extensive experience in optimization algorithms and image analysis techniques.

SOFTWARE

Matlab · Python · Julia · LaTeX · Desmos · Excel